Preparing Training Data for Small Object Detection in Natural Environments

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***Abstract*—Training data quality is a critical but often under-explored factor in developing small object detectors for UAV-based agricultural applications. This paper investigates the effect of systematic dataset curation on olive fruit detection using YOLOv11l, evaluating three preprocessing strategies applied to a 300-image dataset of annotated olive tree canopies. The strategies are: (1) sharpness-based image selection using Laplacian variance and Tenengrad gradient metrics; (2) canopy extraction via monocular depth estimation using the Depth Anything model; and (3) removal of sub-16-pixel bounding box annotations. Applied individually, sharpness-based selection and depth masking offer limited benefit, since global sharpness scores are confounded by non-canopy scene content. When sharpness is evaluated within the canopy mask, Laplacian-based selection produces consistent improvements. Tiny-object removal yields the largest single-method gain, particularly in localisation-sensitive metrics. The fully combined pipeline raises mAP50-95 from 0.594 to 0.659 and mAP50 from 0.832 to 0.873 relative to the uncurated baseline, without any modification to the model or training procedure. These results suggest that for UAV-based precision agriculture, careful dataset preparation can be as impactful as architectural choices.**

***Index Terms*—small object detection, UAV imagery, training data curation, image sharpness, monocular depth estimation, YOLO, olive fruit detection, precision agriculture**

I. Introduction

The detection of small objects in natural environments using deep learning remains a significant challenge in computer vision, particularly in agricultural applications where target objects such as fruits are often minuscule relative to the overall image resolution. To obtain imagery in such cases, unmanned aerial vehicles (UAVs) have become indispensable platforms for agricultural monitoring, offering high-resolution imagery of orchards and plantations [1]–[3]. However, the quality of training data used to develop object detection models is a critical determinant of their ultimate performance. Images captured by UAVs frequently suffer from motion blur caused by platform movement, wind pressure, and camera settings [4], [5], which degrades the feature information necessary for training robust detectors. Furthermore, when target objects are extremely small—such as olive fruits on tree canopies—additional challenges arise from the presence of irrelevant background content and the inherent difficulty that modern detectors such as YOLO face when learning from objects below certain pixel thresholds [6].

This paper addresses the preparation of a high-quality training image database for improving the detection of small objects, specifically olive fruits, in natural environments. Our approach encompasses three complementary strategies: (1) the assessment and filtering of image sharpness using Laplacian and Tenengrad focus measures to discard blurred images from the training set; (2) the exclusion of extremely small object instances (smaller than 16×16 pixels) that may hinder YOLO-based detector training; and (3) the application of monocular depth estimation methods, such as Depth Anything, to segment images by depth and retain only the relevant tree canopy while discarding background elements. By systematically curating the training dataset through these methods, we aim to demonstrate that data quality and relevance significantly influence detection performance for small agricultural objects.

II. Literature Review

A. UAV-Based Agricultural Imaging

UAVs have become a standard platform for agricultural monitoring, providing high-resolution imagery that surpasses traditional satellite-based sensing [3], [7]. In olive cultivation, they have been applied to tree classification, disease detection, canopy segmentation, and health monitoring [1], [2], [8]. Deep learning classifiers trained on UAV imagery have achieved accuracies exceeding 97% for olive tree status classification [1] and 99% precision for disease recognition using transfer learning on a 14,000-image dataset [2]. Canopy segmentation using instance segmentation networks has been shown to enable individual tree biovolume estimation with 82% accuracy, with spatial resolution identified as a key influencing factor [8]. These results confirm the effectiveness of UAV-based imaging for olive tree monitoring, while highlighting that the quality and curation of the captured imagery remain critical factors for downstream detection performance.

B. Image Sharpness Assessment and Blur Filtering

Motion blur caused by platform movement, wind, and camera settings is common in UAV imagery and degrades both annotation quality and detector training [4], [5]. Two classical focus measures are widely used to quantify sharpness. The Laplacian variance detects high-frequency content through second-order derivatives-higher variance indicates a sharper image-and has been applied to motion blur detection in UAV

SLAM [5] and quality-adaptive navigation with an 18.18% accuracy improvement [9]. The Tenengrad method sums squared Sobel gradient responses and has been ranked among the best performers in comprehensive autofocus comparisons across multiple domains [10], [11].

Filtering blurred images before training yields measurable improvements. Applying the Tenengrad function to a UAV tea-bud detection dataset resulted in the removal of 822 blurred images from 2,898, and training on the refined subset produced significantly better detection results [6]. Training exclusively on sharp images has been shown to outperform models trained on blurred data even when deployment conditions include blur [4], and blur removal is consistently described as a key step in UAV preprocessing pipelines [12].

C. Small Object Detection with YOLO

Successive downsampling in YOLO feature extraction reduces the representation of very small objects, making detection unreliable when objects fall below certain pixel thresholds [13]. Dedicated small-object detection heads and multi-scale feature fusion have been proposed to address this limitation, and tiling images into sub-crops to increase the effective relative object size has been shown to improve detection accuracy while reducing labelling errors associated with extremely small bounding boxes [6], [14]. Upsampling low-resolution UAV images by a factor of 4 using Laplacian pyramid super-resolution has similarly been demonstrated to recover sufficient feature information from small objects and improve fault detection performance [15].

D. Depth-Based Background Removal

Background clutter from sky, soil, and surrounding vegetation increases the proportion of irrelevant negative regions in training data. Depth-based segmentation has been shown to effectively isolate relevant objects from background in UAV settings [16], and 3D photogrammetric reconstruction has been used to separate olive canopies from terrain for individual tree characterization [7]. Controlling the spatial relationship between foreground objects and background content in training images has been demonstrated to improve detection reliability in semi-synthetic dataset pipelines [17]. Modern monocular depth estimation models such as Depth Anything extend these approaches to single RGB images, enabling lightweight per-pixel depth inference without stereo cameras or LiDAR.

III. Methodology

In our previous work [18], we highlighted the importance of dataset size and quality on the final performance of the trained network. This study investigates several dataset preparation strategies for improving small object detection in natural agricultural environments. The target task is olive fruit detection using the YOLO object detection framework. Since olive fruits occupy only a small portion of the image and are frequently affected by occlusion, blur, cluttered backgrounds, and scale variation, training dataset preparation techniques are expected to significantly influence detector performance.

A. Dataset

Data were collected in 2025 at an orchard of approximately 5,300 olive trees of different cultivars in Zadar County, Croatia. Imagery was obtained using a DJI Mavic 3 Multispectral UAV with an RGB sensor and narrow-band multispectral sensors. The dataset contains still images of canopies from a height of about 2 metres.

The dataset consists of 300 RGB images of olive trees captured in natural outdoor conditions. To retain original resolution of the tiny objects on a modern GPU, the images were tiled into 16 smaller images containing approximately 19 objects per image. All of the 92,316 visible olive fruits were manually annotated using bounding boxes. The dataset includes significant variation in illumination, focus quality, canopy density, and viewing angle.

For the baseline configuration, the dataset was divided into training, validation, and test subsets using an 80/10/10 split. The same split strategy was maintained across all experiments to ensure comparability between models.

B. Experimental Design

A total of eight dataset preparation strategies were used to train a YOLO model. The experiments were designed to evaluate the influence of:

- image sharpness selection,
- background removal using depth estimation,
- filtering of extremely small annotated objects.

The evaluated configurations are summarised in Table I.

TABLE I
OVERVIEW OF EVALUATED DATASET PREPARATION STRATEGIES.

Method	Dataset Preparation Strategy	Images
M0	Baseline dataset	300
M1a	Laplacian sharpness ranking	200
M1b	Tenengrad sharpness ranking	200
M2	Depth masking	300
M1a+M2	Depth masking + Laplacian ranking	200
M1b+M2	Depth masking + Tenengrad ranking	200
M3	Tiny object filtering	300
M1a+M2+M3	Depth masking + Laplacian + tiny obj. filt.	200

Following our results from [18] we chose YOLOv11l as the benchmark architecture to compare the preparation methods. A trained model for every method was obtained using identical YOLO training parameters to isolate the influence of dataset preparation techniques. The default agumentations include random hue, saturation, and brightness, geometric transformations, channel swapping, Mosaic, MixUp, and CutMix. Due to semantic distribution differences between datasets, the methods are evaluated on their own validation sets.

C. Image Sharpness Ranking

The first approach investigated involved training exclusively on the sharpest images in the dataset, under the assumption that blurred images reduce the visibility of small olive fruits and therefore introduce ambiguity during training. This is particularly important with tiny objects, since the blur size

can become comparable to the size of the object itself, making precise bounding box placement difficult.

Two classical focus measures were evaluated.

1) Laplacian Variance: The Laplacian-based sharpness metric measures the amount of high-frequency image content by computing the variance of the Laplacian operator response. Images with higher variance are considered sharper.

The Laplacian operator is defined as:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \tag{1}$$

For each image, the variance of the Laplacian response was computed and used as a sharpness score. Images were then ranked according to this score, and the top 200 sharpest images were selected for training.

2) Tenengrad Sharpness: The Tenengrad method estimates image sharpness using gradient magnitude obtained from Sobel edge operators. Strong gradients indicate well-focused image regions.

The gradient magnitude is computed as:

$$G = \sqrt{G_x^2 + G_y^2} \tag{2}$$

where G_x and G_y are horizontal and vertical Sobel gradients.

Similarly to the Laplacian experiment, images were ranked according to the Tenengrad score, and the top 200 images were retained for training. Figure 1 contains the comparison between the best and worst ranked image from the dataset.

D. Background Removal Using Depth Estimation

Natural orchard scenes contain complex backgrounds including sky, soil, agricultural equipment, and neighbouring vegetation. Since olive fruits appear only within the tree canopy, reducing irrelevant image regions may help the detector focus on informative features.

To achieve this, a depth-based masking pipeline was developed. The process consists of three stages:

1) monocular depth estimation using the Depth Anything model,
2) foreground extraction using depth thresholding (a threshold value of 60 out of 255 was chosen based on a subset of images to remove objects more distant than the canopy),
3) ground removal using column-wise depth gradient analysis with moving-average smoothing.

This results in a binary mask preserving primarily the olive tree canopy while suppressing distant background regions. The mask was then applied to the original RGB image before training.

An additional motivation for combining depth masking with sharpness ranking was the observation that sharp non-canopy regions (e.g., background objects, soil, or nearby branches) can artificially increase global sharpness scores. By restricting the image content to the canopy area before ranking, the sharpness metrics become more representative of the actual fruit regions.

Figure 2 illustrates the canopy extraction procedure. The original image (left) is passed through the Depth Anything model to produce a depth map (centre); the canopy is then extracted by depth thresholding and ground removal (right).

E. Tiny Object Filtering

The final investigated preprocessing method addressed extremely small annotated objects. Very small bounding boxes frequently correspond to distant fruits, partially occluded objects, or annotation uncertainty. Such objects may contribute limited useful information during training while increasing localisation difficulty.

To reduce annotation noise, objects whose larger bounding box dimension was smaller than 16 pixels were removed from the annotations.

Formally, an object was retained only if:

$$\max(w, h) \geq 16 \tag{3}$$

where w and h denote the width and height of the bounding box in pixels.

This filtering strategy removed 13,514 out of 92,613 total annotated objects. It was evaluated both independently and in combination with the best-performing preprocessing pipeline.

F. Final Combined Configuration

The final experiment combined all preprocessing strategies that were hypothesized to improve small object visibility:

- Laplacian-based sharpness ranking,
- depth-based canopy masking,
- tiny object filtering.

This configuration aimed to produce a cleaner training dataset containing:

- visually sharp canopy regions,
- reduced background clutter,
- more reliable object annotations.

The resulting trained model was compared against the baseline configuration to evaluate the cumulative effect of the proposed preprocessing pipeline.

IV. Results

A. Training Setup

All models were trained on a workstation equipped with an NVIDIA GeForce RTX 5060 Ti (16 GB VRAM), an AMD Ryzen 9 7900X processor, and 32 GB of RAM, using PyTorch with CUDA/cuDNN acceleration and automatic mixed precision (AMP).

All experiments used the YOLOv11l model variant and identical hyperparameters to isolate the effects of dataset preparation. Training ran for up to 300 epochs with early stopping at a patience of 100 epochs, monitoring validation mAP50-95; the best checkpoint per run was retained. The batch size was set to 4, with an AdamW optimizer (learning rate 10^{-3}) or SGD with momentum 0.937 (learning rate 10^{-2}). Input images were tiled into 16 tiles of 1320×989 pixels to fit within GPU memory. The reported results present single

Fig. 1. The sharpest image (left) and the least sharp image (right) according to the Laplacian metric.

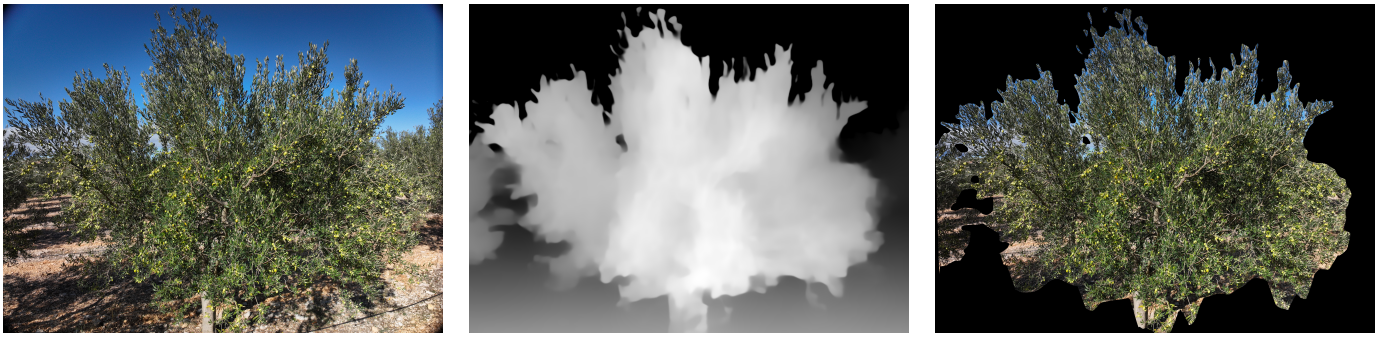

Fig. 2. Canopy extraction pipeline: original image (left), depth mask (centre), and final image (right).

training run, as computational resource constraints limited repeated runs.

Performance is reported in terms of precision (P), recall (R), mAP50, and mAP50-95 evaluated on the per-dataset validation split. Because each experimental variant is evaluated on its own validation set (derived from the correspondingly filtered dataset), the metrics are not directly comparable in absolute terms; they nonetheless provide a consistent relative signal for assessing each preprocessing strategy.

B. Effect of Sharpness-Based Image Selection

Table II compares the baseline model M0 against variants M1a (Laplacian-based ranking) and M1b (Tenengrad-based ranking), in which training is restricted to the sharpest images according to each metric.

TABLE II
EFFECT OF SHARPNESS-BASED IMAGE SELECTION ON DETECTION PERFORMANCE.

Method	P	R	mAP50	mAP50-95
M0 (baseline)	0.823	0.738	0.832	0.594
M1a (Laplacian)	0.822	0.747	0.833	0.610
M1b (Tenengrad)	0.838	0.719	0.820	0.568

Neither sharpness metric yields a consistent improvement over the baseline in isolation. This is likely because high-frequency content in these images is not exclusive to the fruit-bearing canopy: ground, vehicles, and other scene elements also contribute to sharpness scores, introducing noise into the selection process.

C. Effect of Canopy Extraction

Table III evaluates M2, in which a depth-based mask is applied to retain only the canopy region before training.

TABLE III
EFFECT OF CANOPY EXTRACTION VIA DEPTH MASKING ON DETECTION PERFORMANCE.

Method	P	R	mAP50	mAP50-95
M0 (baseline)	0.823	0.738	0.832	0.594
M2 (canopy)	0.831	0.730	0.826	0.589

Depth masking alone provides no meaningful gain. This is consistent with the dataset characteristics: the vast majority of annotated fruits are located on the canopy, so masking out other regions does not substantially change the distribution of training examples.

D. Combining Canopy Extraction with Sharpness Selection

Table IV reports the combined methods M1a+M2 and M1b+M2, where sharpness is evaluated exclusively within the canopy mask before image selection.

TABLE IV
EFFECT OF COMBINING CANOPY EXTRACTION WITH SHARPNESS-BASED SELECTION.

Method	P	R	mAP50	mAP50-95
M0 (baseline)	0.823	0.738	0.832	0.594
M1a+M2	0.845	0.750	0.847	0.618
M1b+M2	0.832	0.741	0.835	0.604

Restricting sharpness evaluation to the canopy region yields clear improvements over both the baseline and the individual methods. The Laplacian-based variant (M1a+M2) achieves gains across all four metrics, confirming the earlier hypothesis: global sharpness scores are confounded by non-canopy scene content, and masking them out allows the metric to target the regions that are actually relevant to the detection task.

E. Effect of Tiny-Object Removal

Table V shows the effect of removing annotations smaller than 16×16 pixels (M3).

TABLE V
EFFECT OF TINY-OBJECT REMOVAL ON DETECTION PERFORMANCE.

Method	P	R	mAP50	mAP50-95
M0 (baseline)	0.823	0.738	0.832	0.594
M3 (no tiny objects)	0.830	0.776	0.867	0.639

Filtering sub-16-pixel objects produces a substantial improvement, particularly in recall and mAP50-95. This is expected behaviour: YOLO's multi-scale feature extraction progressively downsamples the input, causing spatial information from very small objects to become diffuse before reaching the detection heads. Including such annotations during training introduces an ambiguous supervision signal; removing them produces cleaner gradients and a more consistent evaluation target.

F. Combined Pipeline

Table VI presents the fully combined pipeline M1a+M2+M3, applying canopy-masked Laplacian selection together with tiny-object removal.

TABLE VI
COMBINED PIPELINE VS. THE BASELINE.

Method	P	R	mAP50	mAP50-95
M0 (baseline)	0.823	0.738	0.832	0.594
M1a+M2+M3	0.840	0.784	0.873	0.659

The combined pipeline achieves the best results across all metrics, with mAP50-95 improving from 0.594 to 0.659 (+6.5%) and mAP50 from 0.832 to 0.873 (+4.1%) relative to the baseline. The pipeline consists of three sequential steps: (1) canopy extraction via depth masking, (2) selection of the top-200 images ranked by within-canopy Laplacian sharpness, and (3) removal of sub-16-pixel object annotations.

V. DISCUSSION

The results demonstrate that targeted dataset curation can yield meaningful improvements in aerial fruit detection without any changes to the model architecture or training regime. Three observations are worth highlighting.

First, sharpness-based image selection is only effective when evaluated within a semantically relevant region. Applied globally, both the Laplacian and Tenengrad metrics are confounded by non-canopy content, and neither improves over the baseline. Once the sharpness score is restricted to the canopy mask, the Laplacian variant produces consistent gains. This suggests that for UAV-acquired orchard imagery, sharpness alone is an insufficient signal for data quality unless it is grounded in scene understanding.

Second, tiny-object removal has a disproportionately large effect on mAP50-95, the stricter localisation-sensitive metric. This is consistent with the known difficulty of detecting sub-resolution objects in single-stage detectors: annotations below the effective receptive field of the network's detection heads act as noise during training. Removing them not only cleans up evaluation but also produces a more learnable supervision signal.

Third, the two strategies are complementary. The sharpness-based selection improves the quality of images presented to the network, while tiny-object filtering improves the quality of labels. Together they yield the best performance, suggesting that both image-level and annotation-level curation matter independently. Future work could explore learned sharpness proxies, adaptive size thresholds based on network architecture, and the interaction of these filters with data augmentation strategies.

VI. CONCLUSION

This paper investigated the effect of dataset curation on aerial fruit detection with YOLOv11l, examining three preprocessing strategies: sharpness-based image selection, canopy extraction via depth masking, and removal of sub-pixel-threshold annotations. Applied individually, each strategy offers limited or inconsistent benefit. However, combining canopy-masked Laplacian sharpness ranking with tiny-object filtering yields a consistent improvement over the baseline across all metrics, raising mAP50-95 from 0.594 to 0.659 on method specific validation sets without any modification to the model or training procedure.

The key finding is that data quality is both image-level and annotation-level: selecting sharper views of the canopy and removing ambiguous supervision signal are complementary interventions. This suggests that for UAV-based precision agriculture applications, careful dataset preparation may be as impactful as architectural choices, and should be treated as a first-class component of the model development pipeline.

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